

Product Service Documentation

Overview

The Product Service manages the bookstore's product catalog. It handles product creation, retrieval, searching, filtering, and categorization. Products can be browsed by customers and managed by the owner.

Service Responsibilities

- Product CRUD operations
- Product search and filtering
- Category-based product organization
- Board-based filtering (SSC, HSC, ICSE, CBSE)
- Class/grade filtering
- Featured, bestseller, and trending product management
- Product image management

API Endpoints

1. List Products

Endpoint: GET /api/products

Description: Get paginated list of products with optional filters

Query Parameters:

- `page` (int, default: 0): Page number
- `size` (int, default: 20): Page size
- `categoryId` (Long, optional): Filter by category
- `board` (String, optional): Filter by board (SSC, HSC, ICSE, CBSE)
- `class` (String, optional): Filter by class/grade
- `search` (String, optional): Search by name or ISBN
- `featured` (Boolean, optional): Filter featured products

- **bestseller** (Boolean, optional): Filter bestseller products
- **sort** (String, optional): Sort field (name, price, createdAt)
- **direction** (String, optional): Sort direction (ASC, DESC)

Response: 200 OK

```
{
  "content": [
    {
      "id": 1,
      "name": "Mathematics for Class 10",
      "description": "Comprehensive mathematics textbook for SSC board",
      "isbn": "978-1234567890",
      "author": "Dr. Ramesh Kumar",
      "publisher": "Educational Publishers",
      "price": 450.00,
      "categoryId": 1,
      "categoryName": "School Books",
      "board": "SSC",
      "class": "10",
      "featured": true,
      "bestseller": false,
      "imageUrl": "/images/products/math-10.jpg",
      "createdAt": "2025-12-23T10:00:00Z"
    }
  ],
  "page": 0,
  "size": 20,
  "totalElements": 150,
  "totalPages": 8
}
```

Authorization: Public (no authentication required)

2. Get Product Details

Endpoint: GET /api/products/{id}

Description: Get detailed information about a specific product

Path Parameters:

- **id** (Long): Product ID

Response: 200 OK

```
{
  "id": 1,
  "name": "Mathematics for Class 10",
  "description": "Comprehensive mathematics textbook for SSC board covering all topics...",
  "isbn": "978-1234567890",
  "author": "Dr. Ramesh Kumar",
  "publisher": "Educational Publishers",
  "price": 450.00,
  "categoryId": 1,
  "categoryName": "School Books",
  "categoryPath": "School Books > SSC > Class 10",
  "board": "SSC",
  "class": "10",
  "featured": true,
  "bestseller": false,
  "imageUrl": "/images/products/math-10.jpg",
  "createdAt": "2025-12-23T10:00:00Z",
  "updatedAt": "2025-12-23T10:00:00Z"
}
```

Error Responses:

- 404 Not Found : Product not found

Authorization: Public

3. Search Products

Endpoint: GET /api/products/search

Description: Search products by name, ISBN, author, or publisher

Query Parameters:

- q (String, required): Search query
- page (int, default: 0): Page number
- size (int, default: 20): Page size

Response: 200 OK

```
{
  "content": [
    {
      "id": 1,
      "name": "Mathematics for Class 10",
      "isbn": "978-1234567890",
      "author": "Dr. Ramesh Kumar",
      "price": 450.00,
      "imageUrl": "/images/products/math-10.jpg"
    }
  ],
  "page": 0,
  "size": 20,
  "totalElements": 5
}
```

Authorization: Public

4. Get Products by Category

Endpoint: GET /api/products/category/{categoryId}

Description: Get all products in a specific category

Path Parameters:

- categoryId (Long): Category ID

Query Parameters:

- page (int, default: 0): Page number
- size (int, default: 20): Page size

Response: 200 OK (Same structure as List Products)

Authorization: Public

5. Get Products by Board

Endpoint: GET /api/products/board/{board}

Description: Get products filtered by educational board

Path Parameters:

- `board` (String): Board name (SSC, HSC, ICSE, CBSE)

Query Parameters:

- `page` (int, default: 0): Page number
- `size` (int, default: 20): Page size

Response: 200 OK (Same structure as List Products)

Authorization: Public

Valid Board Values: SSC, HSC, ICSE, CBSE

6. Get Products by Class

Endpoint: GET `/api/products/class/{class}`

Description: Get products filtered by class/grade

Path Parameters:

- `class` (String): Class/grade (e.g., "10", "12", "First Year")

Query Parameters:

- `page` (int, default: 0): Page number
- `size` (int, default: 20): Page size

Response: 200 OK (Same structure as List Products)

Authorization: Public

7. Create Product

Endpoint: POST `/api/products`

Description: Create a new product (owner only)

Headers: Authorization: Bearer {token}

Request Body:

```
{
  "name": "Physics for Class 12",
  "description": "Advanced physics textbook for HSC board",
  "isbn": "978-1234567891",
  "author": "Dr. Suresh Patel",
  "publisher": "Science Publishers",
  "price": 550.00,
  "categoryId": 1,
  "board": "HSC",
  "class": "12",
  "featured": false,
  "bestseller": true,
  "imageUrl": "/images/products/physics-12.jpg"
}
```

Response: 201 Created

```
{
  "id": 2,
  "name": "Physics for Class 12",
  "description": "Advanced physics textbook for HSC board",
  "isbn": "978-1234567891",
  "author": "Dr. Suresh Patel",
  "publisher": "Science Publishers",
  "price": 550.00,
  "categoryId": 1,
  "board": "HSC",
  "class": "12",
  "featured": false,
  "bestseller": true,
  "imageUrl": "/images/products/physics-12.jpg",
  "createdAt": "2025-12-23T11:00:00Z"
}
```

Authorization: OWNER only

Validation Rules:

- Name is required
- ISBN must be unique
- Price must be positive

- Category must exist

8. Update Product

Endpoint: PUT /api/products/{id}

Description: Update an existing product (owner only)

Headers: Authorization: Bearer {token}

Path Parameters:

- id (Long): Product ID

Request Body: (Same as Create Product, all fields optional)

Response: 200 OK (Updated product)

Authorization: OWNER only

Error Responses:

- 404 Not Found : Product not found

9. Delete Product

Endpoint: DELETE /api/products/{id}

Description: Delete a product (owner only)

Headers: Authorization: Bearer {token}

Path Parameters:

- id (Long): Product ID

Response: 204 No Content

Authorization: OWNER only

Error Responses:

- 404 Not Found : Product not found
- 409 Conflict : Product has associated inquiries

10. Get Featured Products

Endpoint: GET /api/products/featured

Description: Get all featured products

Query Parameters:

- `limit` (int, default: 10): Maximum number of products

Response: 200 OK

```
[
  {
    "id": 1,
    "name": "Mathematics for Class 10",
    "price": 450.00,
    "imageUrl": "/images/products/math-10.jpg"
  }
]
```

Authorization: Public

11. Get Bestseller Products

Endpoint: GET /api/products/bestsellers

Description: Get all bestseller products

Query Parameters:

- `limit` (int, default: 10): Maximum number of products

Response: 200 OK (Same structure as Featured Products)

Authorization: Public

12. Get Trending Products

Endpoint: GET /api/products/trending

Description: Get trending products (based on inquiry count or views)

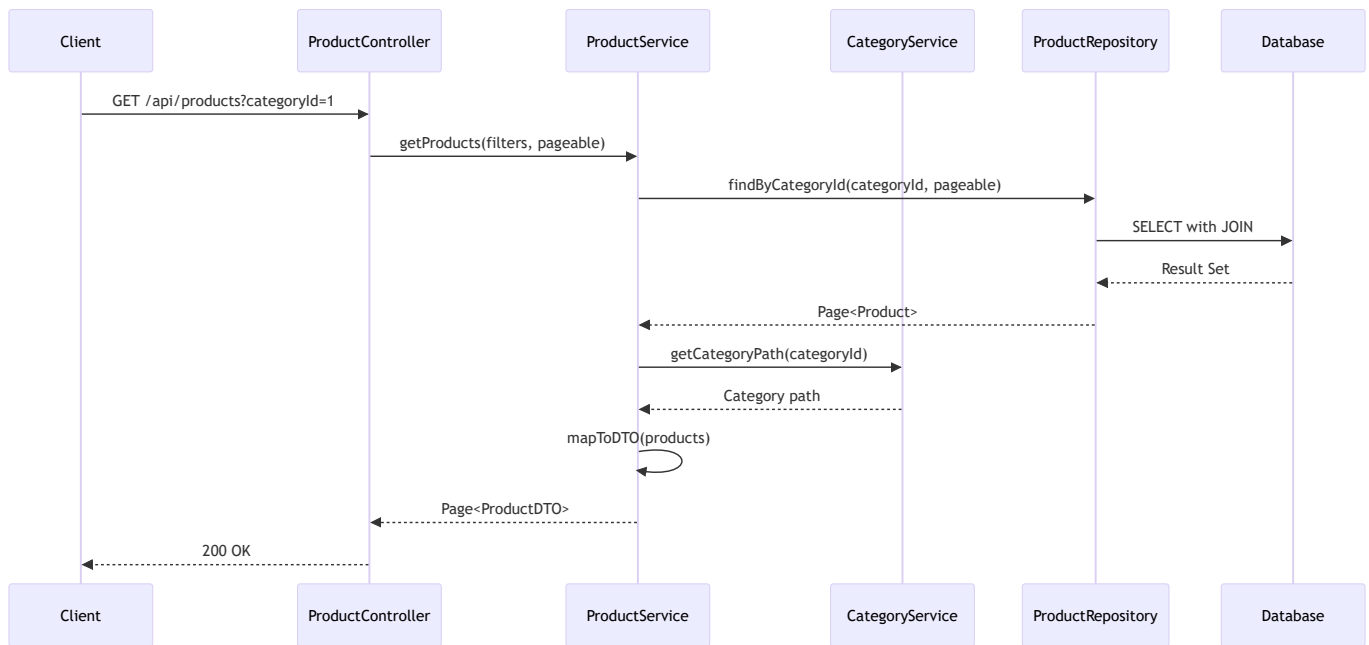
Query Parameters:

- `limit` (int, default: 10): Maximum number of products
- `period` (String, optional): Time period (WEEK, MONTH, YEAR)

Response: 200 OK (Same structure as Featured Products)

Authorization: Public

Service Flow



Product Entity

```
@Entity
@Table(name = "products")
public class Product {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(nullable = false)
    private String name;

    @Column(columnDefinition = "TEXT")
    private String description;

    @Column(unique = true)
    private String isbn;

    private String author;
    private String publisher;

    @Column(nullable = false)
    private BigDecimal price;

    @ManyToOne
    @JoinColumn(name = "category_id")
    private Category category;

    private String board; // SSC, HSC, ICSE, CBSE
    private String class; // Class/Grade

    private Boolean featured;
    private Boolean bestseller;

    private String imageUrl;

    @CreationTimestamp
    private LocalDateTime createdAt;

    @UpdateTimestamp
    private LocalDateTime updatedAt;
}
```

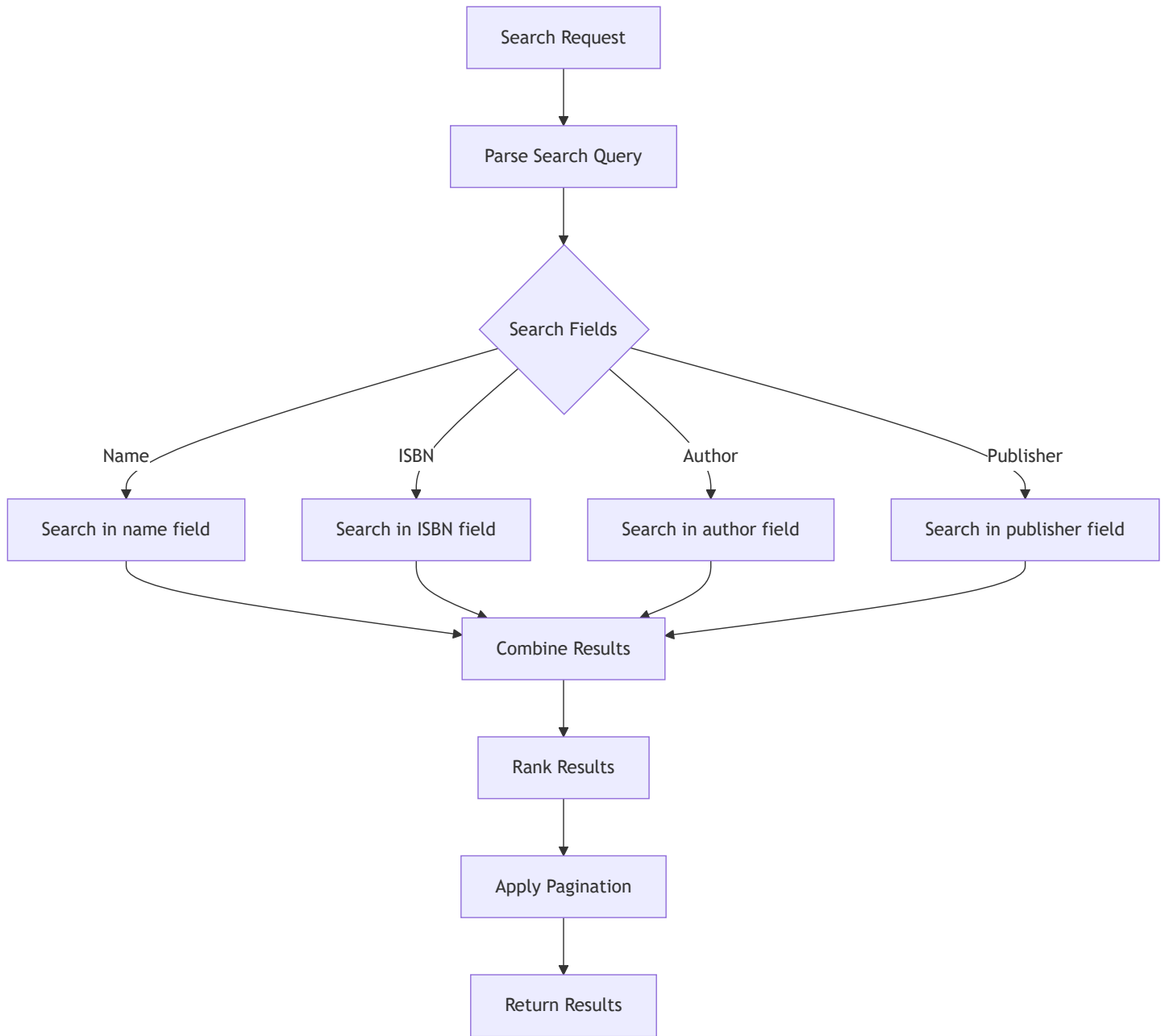
Service Methods

ProductService Interface

```
public interface ProductService {  
    Page<ProductDTO> getProducts(ProductFilter filter, Pageable pageable);  
    ProductDTO getProductById(Long id);  
    Page<ProductDTO> searchProducts(String query, Pageable pageable);  
    Page<ProductDTO> getProductsByCategory(Long categoryId, Pageable pageable);  
    Page<ProductDTO> getProductsByBoard(String board, Pageable pageable);  
    Page<ProductDTO> getProductsByClass(String className, Pageable pageable);  
    ProductDTO createProduct(CreateProductRequest request);  
    ProductDTO updateProduct(Long id, UpdateProductRequest request);  
    void deleteProduct(Long id);  
    List<ProductDTO> getFeaturedProducts(int limit);  
    List<ProductDTO> getBestsellerProducts(int limit);  
    List<ProductDTO> getTrendingProducts(int limit, String period);  
}
```

Search and Filtering Logic

Search Implementation



Filtering Strategy

- **Category Filter:** Uses JOIN with categories table
- **Board Filter:** Direct WHERE clause on board column
- **Class Filter:** Direct WHERE clause on class column
- **Combined Filters:** Uses AND conditions for multiple filters
- **Indexing:** Indexes on categoryId, board, class for performance

Database Queries

Optimized Query Example

```
SELECT p.*, c.name as category_name
FROM products p
LEFT JOIN categories c ON p.category_id = c.id
WHERE
    (:categoryId IS NULL OR p.category_id = :categoryId)
    AND (:board IS NULL OR p.board = :board)
    AND (:class IS NULL OR p.class = :class)
    AND (:search IS NULL OR
        p.name LIKE %:search% OR
        p.isbn LIKE %:search% OR
        p.author LIKE %:search%)
ORDER BY p.created_at DESC
LIMIT :size OFFSET :offset
```

Performance Considerations

Caching Strategy

- **Product List Cache:** Cache frequently accessed product lists
- **Product Detail Cache:** Cache individual product details
- **Category Products Cache:** Cache products by category
- **Cache Invalidation:** Invalidate on product create/update/delete

Indexing

- **Primary Key:** id (auto-indexed)
- **Unique Index:** isbn
- **Indexes:** categoryId, board, class, featured, bestseller
- **Full-Text Index:** name, description (for search)

Pagination

- All list endpoints support pagination
- Default page size: 20

- Maximum page size: 100
- Efficient offset-based pagination

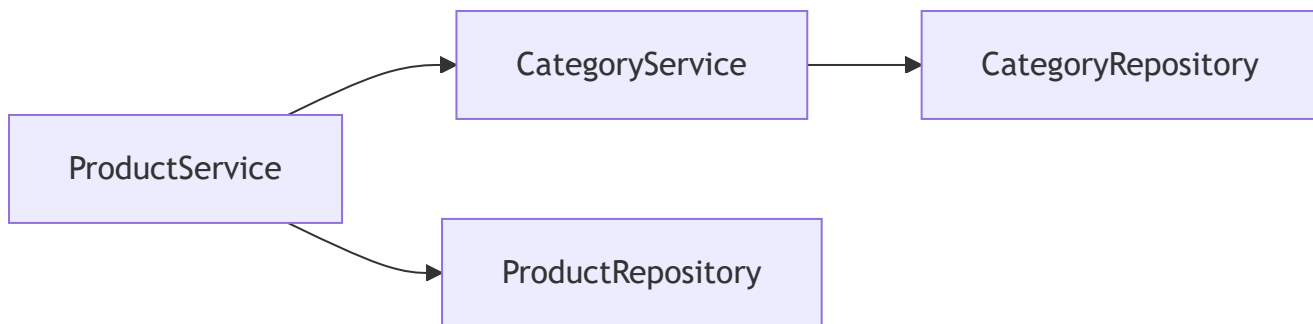
Error Handling

Custom Exceptions

- **ProductNotFoundException**: When product is not found
- **ISBNAlreadyExistsException**: When ISBN is duplicate
- **CategoryNotFoundException**: When category doesn't exist
- **InvalidProductDataException**: When validation fails

Integration with Other Services

Category Service Integration



- Product Service calls Category Service to validate category existence
- Category Service provides category path for product display

Testing Considerations

Unit Tests

- Product creation with valid/invalid data
- Search functionality
- Filtering logic
- Pagination

Integration Tests

- Complete CRUD operations
- Search with multiple criteria
- Filter combinations
- Category validation

Future Enhancements

- Product reviews and ratings
- Product recommendations
- Inventory tracking (if needed)
- Bulk product import
- Product variants (editions, formats)
- Advanced search with facets